

From the Economic System to a Habitable Planet

One renewable energy mix simulated under four economic systems, then tested against the nine planetary boundaries - from the capitalist market to the RCED, by way of a social-democratic variant

Central thesis: the binding constraint on the transition is institutional, not technical. The same hardware can decarbonize or not, leanly or wastefully - and decarbonizing energy is not enough: food and materials must also be reformed to stay within a habitable planet.

Executive Summary

This study takes a single physical energy system - a fully electrified United States running on solar PV, wind, CSP, solar thermal, with a seasonal sand battery and residential batteries - and applies four different economic frameworks to it, holding the demand for services identical. The results separate cleanly along two axes: carbon and resource use.

The **capitalist market** is capital-lean but leaves climate largely unsolved (~1011 Mt CO₂/yr), because it finances neither seasonal storage nor heat networks. **Soviet** planning decarbonizes almost completely (~20 Mt) but at massive waste (+48% energy, 8550 GW installed). The **RCED** alone reaches the ideal corner - low carbon and lean. A **social-democratic variant** (private property kept) captures ~80% of the RCED's climate benefit (~209 Mt) by building the infrastructure through public investment.

But energy is only an entry point. As of 2025, seven of the nine planetary boundaries are transgressed; decarbonizing energy addresses only two. Adding **food reform** (agroecology, less meat) and then **plastic limitation**, the social-democratic model ends up addressing all seven transgressed boundaries - at a level of political acceptability far higher than the radical RCED. The only remaining structural difference is the unresolved question of material sufficiency versus growth.

1. Question and method

All simulations rest on an 8760-hour hourly-dispatch model: synthetic seasonal and diurnal solar and wind profiles, demand decomposed (low- and high-temperature heat, specific electricity, electrified transport, cooling), and a storage cascade following the principle of energy subsidiarity (batteries for the daily cycle, sand for the seasonal cycle, the national grid for spatial smoothing). The demand for *services* is held identical across systems, so that only the institutions vary. The planetary-boundary analysis then draws on the 2025 scientific assessment (Planetary Health Check) and on the EAT-Lancet Commission 2.0 for the food dimension.

2. The same mix under four economic systems

The central result is that technology does not decide the outcome - institutions do. Each system fails, or succeeds, on a distinct axis. The market under-invests in non-marketable infrastructure and stays tied to gas. The USSR builds the infrastructure and decarbonizes, but by brute force, wasting energy and materials. The RCED combines the market's efficiency with the plan's ability to build the non-marketable, adding demand discipline.

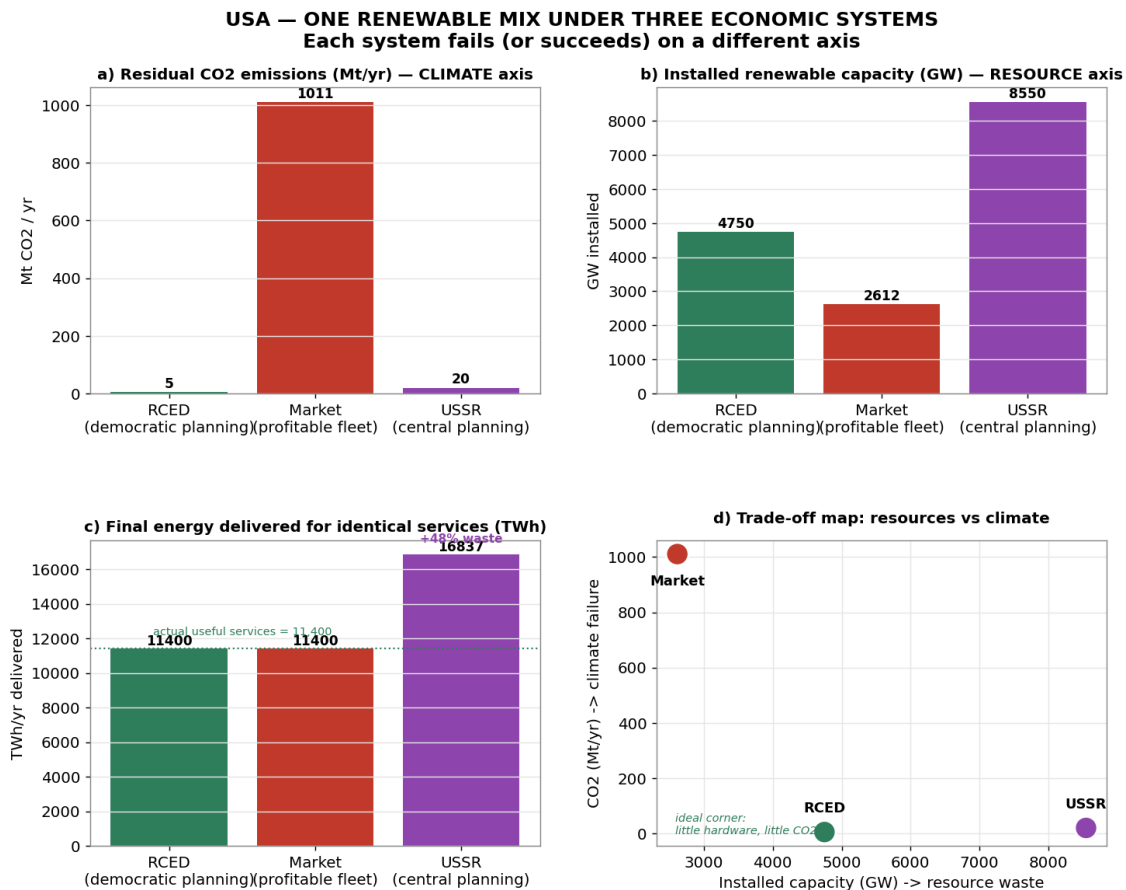


Figure 1 - Three systems, one mix. (a) emissions; (b) installed capacity; (c) energy delivered for identical services; (d) the trade-off map, with the RCED alone in the ideal low-carbon / lean corner.

2.1 The social-democratic variant

To gain acceptability, private property and markets are kept, but the state and municipalities build the network infrastructure (national grid, urban heat networks, public seasonal storage - the Danish model), with carbon pricing. The result is striking: fossil heat falls from 70% to 17%, and CO₂ from ~1011 to ~209 Mt, **without abolishing private property**. The binding constraint was therefore not the property regime, but public investment in the infrastructure the market will not finance.

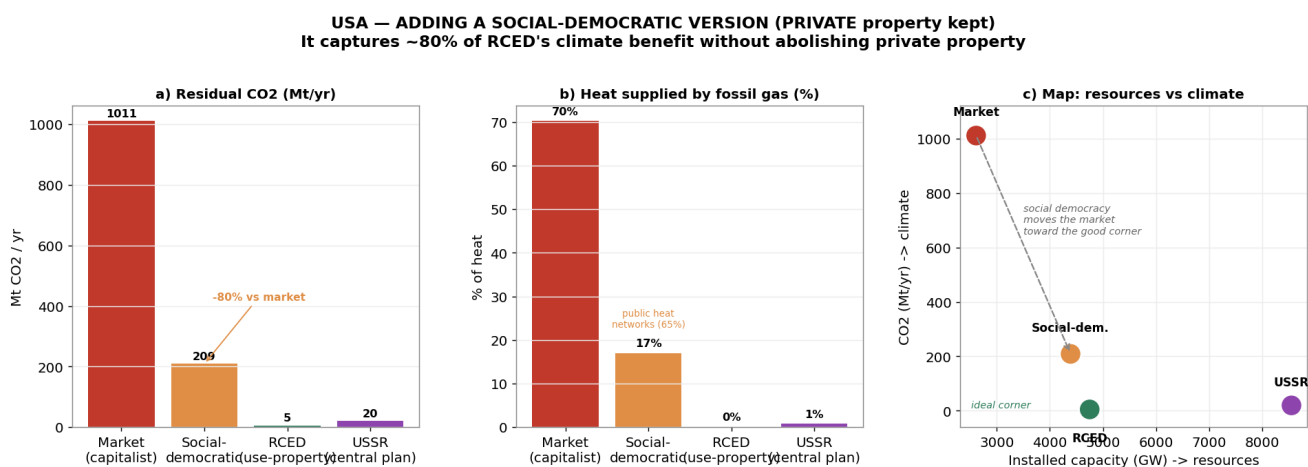


Figure 2 - Adding the social-democratic variant. It captures ~80% of the RCED's climate benefit while keeping private property, and sits near the ideal corner (right-hand map).

2.2 Energy summary

Indicator	Market	Social-dem.	RCED	USSR
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Residual CO ₂ (Mt/yr)	1011	209	~5	20
Heat from gas	70%	17%	0%	1%
Renewable electricity	84%	100%	100%	100%
Installed capacity (GW)	2,612	4390	4,750	8550
Energy / same services	ref.	+11%	ref.	+48%

2.3 The residential-battery question

A common objection: why doesn't the market rescue itself with residential batteries? The sweep shows the limit: even equipping every US home, emissions fall only by a few percent, then plateau. Short-duration batteries shift midday solar to the evening - the daily cycle - but stay empty through multi-day winter lulls and cannot store heat. In the complete system (with seasonal sand), they do find their place: they run the daily cycle (~150 cycles/yr) while the sand handles the season (~1 cycle/yr). Complementary tools, not competitors.

3. Beyond energy: the nine planetary boundaries

As of 2025, seven of the nine planetary boundaries are transgressed (climate, biosphere integrity, land-system change, freshwater, nitrogen/phosphorus flows, novel entities, ocean acidification); only ozone and aerosols remain in the safe zone. Decarbonizing energy meaningfully acts on just two of them, and building the fleet itself adds pressure on land and novel entities (mining, materials). Necessary, but far from sufficient.

The social-democratic model against the 9 PLANETARY BOUNDARIES (2025: 7/9 transgressed)
Decarbonizing energy touches only 2 of the 7 transgressed boundaries - necessary but far from sufficient

Planetary boundary	2025 status	Main driver	Effect of socdem model
Climate change	transgressed	fossil energy + land + agriculture	 partial
Ocean acidification	transgressed (2025) CO ₂ (energy-linked)		 partial
Biosphere integrity	transgressed	land use, agriculture, habitats	 not addressed
Land-system change	transgressed	agriculture, urbanization, mining	 added pressure
Freshwater	transgressed	agriculture, industry	 not addressed
Biogeochemical flows (N,P)	transgressed	fertilizer / industrial agriculture	 not addressed
Novel entities (plastics, chemicals)	transgressed	industrial production/consumption	 added pressure
Atmospheric aerosols	safe (rising)	combustion	 helps (less combustion)
Stratospheric ozone	safe	CFCs (resolved)	 n/a

 net help
  partial help
  not addressed / neutral
  ADDED pressure (materials, mining, land)

Figure 3 - The energy model against the 9 boundaries: it addresses only 2 of the 7 transgressed, and worsens two (land, novel entities) through materials demand.

3.1 Food reform - the missing lever

Food systems are the leading driver of boundary transgression: ~30% of greenhouse gases, almost the entirety of the nitrogen/phosphorus overshoot, the top driver of biodiversity collapse (via land conversion), ~70% of freshwater. Replacing industrial agriculture with agroecology and sharply reducing meat acts precisely on the five boundaries energy cannot touch. The synergy is essential: cutting meat frees the land that makes agroecology (with its lower yields) land-compatible. A major health co-benefit (up to ~15

million deaths averted per year) reinforces acceptability.

Social-democratic model + FOOD REFORM (agroecology, less meat, plant-based diet)
Food is the #1 driver of the transgressed boundaries: it acts where energy cannot

Planetary boundary (2025)	Main driver	Energy alone	+ Food
Climate change <i>transgressed</i>	energy + land + agriculture	partial	decisive
Ocean acidification <i>transgressed (2025)</i>	CO2	partial	partial
Biosphere integrity <i>transgressed</i>	land conversion, agriculture	-	decisive
Land-system change <i>transgressed</i>	agriculture (esp. livestock)	pressure +	decisive
Freshwater <i>transgressed</i>	agriculture (~70% of water)	-	decisive
Biogeochemical flows (N,P) <i>transgressed</i>	fertilizer - almost 100% food	-	decisive
Novel entities <i>transgressed</i>	pesticides + plastics + chemicals	pressure +	partial
Atmospheric aerosols <i>safe</i>	combustion	decisive	decisive
Stratospheric ozone <i>safe</i>	CFCs (resolved)	-	-

■ decisive help ■ partial help ■ not addressed / n/a ■ added pressure
 Bottom line: from 2 transgressed boundaries addressed (energy alone) to 6 of 7 (energy + food). Mainly industrial novel entities remain.

Figure 4 - Energy alone vs energy + food reform. Food moves the tally from 2 to 6 of the 7 transgressed boundaries actually addressed.

4. Plastics and the final comparison: social democracy vs RCED

One transgressed boundary remained unaddressed by the social-democratic path: novel entities (plastics, chemicals). But plastic limitation is achieved through regulation (bans, deposit schemes, extended producer responsibility, REACH-style chemical rules) - without touching private property. Adding it, the social-democratic model now addresses all seven transgressed boundaries, exactly like the full RCED, and enters the 'habitable planet' zone at far higher acceptability.

Social-dem + food + PLASTIC LIMITATION vs full RCED
With plastics, both models equalize on all 7 transgressed boundaries - only ONE difference remains

Boundary / dimension	Social-dem + food + plastics	Full RCED
Climate change	decisive	decisive
Ocean acidification	partial	partial
Biosphere integrity	decisive	decisive
Land-system change	decisive	decisive
Freshwater	decisive	decisive
Biogeochemical flows (N,P)	decisive	decisive
Novel entities (plastics, chemicals)	✓ decisive	decisive
Material sufficiency / post-growth	✗ not addr.	decisive

All 7 transgressed boundaries: identical. Only the last row (growth/sufficiency) still separates the two.

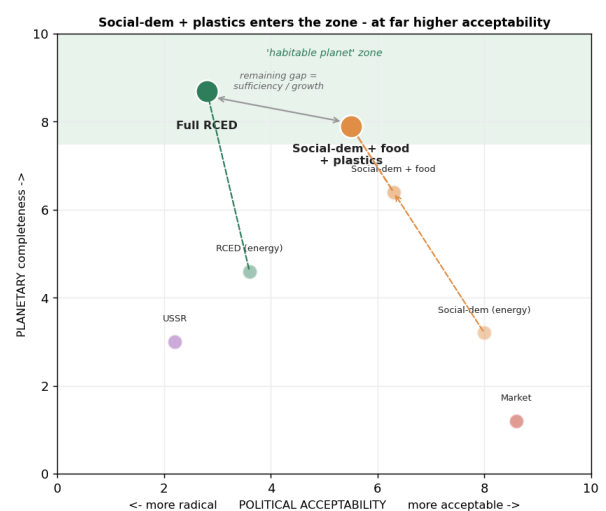


Figure 5 - With plastic limitation, the social-democratic model equalizes the RCED on all 7 transgressed boundaries. Only the 'material sufficiency / growth' dimension remains (right map: it enters the zone, more accessible than the RCED).

The only remaining difference is no longer a particular boundary, but a question of **method** and **durability**. The social-democratic model reaches the zone through regulation, each boundary 'patched' by a dedicated policy bolted onto an intact growth engine. The RCED holds the zone through structure:

use-property removes the rent incentive and thus the driver of overproduction. If absolute material decoupling holds (the green-growth bet), the social-democratic model stays in the zone; if not (the post-growth thesis), it risks drifting back out over time, whereas the RCED's structure holds it there more robustly.

5. Conclusions

- **Institutions decide, not technology.** The same renewable mix can leave ~1 Gt of CO₂ (market) or decarbonize almost completely (RCED, USSR, social-democratic). What changes is who finances the non-marketable infrastructure - seasonal storage, heat networks, national grid.
- **Private property is not the climate constraint.** The social-democratic variant captures most of the RCED's benefit while keeping private property: it suffices to invest publicly in infrastructure and price carbon.
- **Energy is not enough.** Seven of the nine planetary boundaries are transgressed; energy addresses only two. Food (agroecology, less meat) addresses five more; plastic limitation closes the seventh.
- **A viable, complete model exists.** Social democracy + energy decarbonization + food reform + plastic limitation addresses all seven transgressed boundaries while remaining markedly more acceptable than the radical RCED.
- **The only genuinely open question is growth.** The ultimate difference between the two models reduces to the bet on material decoupling: is regulation enough to rein in material throughput, or must its driver be removed structurally? This is the green-growth / post-growth debate, scientifically unresolved - with the evidence leaning toward 'regulation alone struggles on materials'.

6. Limitations

- Deterministic simulations, a single representative year, synthetic weather, a single national node (no inter-regional congestion); gas is treated as unlimited flexibility.
- The Soviet mix is held renewable for comparability; a historical plan would have centralized on coal and nuclear, with higher emissions - the ~20 Mt is a favorable floor.
- The planetary-boundary scorecards are qualitative judgments grounded in the literature (EAT-Lancet 2.0; Planetary Health Check 2025), not outputs of a single integrated model.
- The analysis covers the United States (with Switzerland as reference); the global scale reopens the governance question (RCED Part II), beyond this scope.

Main sources: Planetary Health Check 2025 (Stockholm Resilience Centre / PIK); EAT-Lancet Commission 2.0 (2025); Vogel & Hickel, Lancet Planetary Health (2023); RCED model v5.0.1 (N. Felder). Simulations and figures produced for this report.